

The application of IR spectroscopy in fish aging: pros and cons of the method

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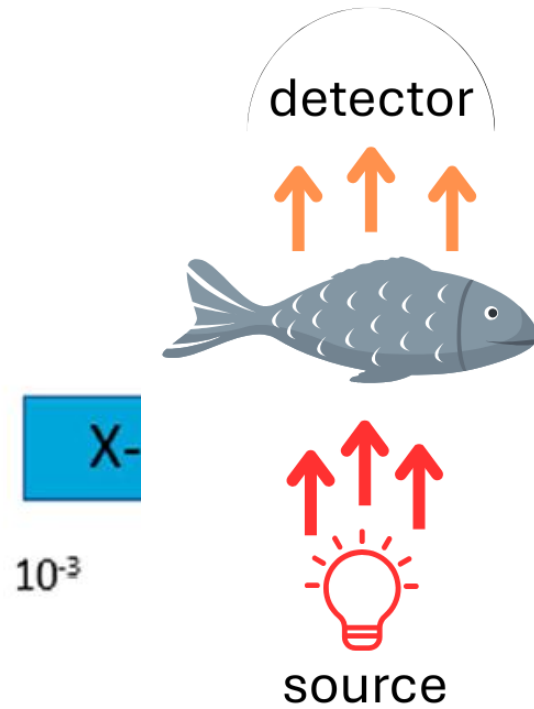
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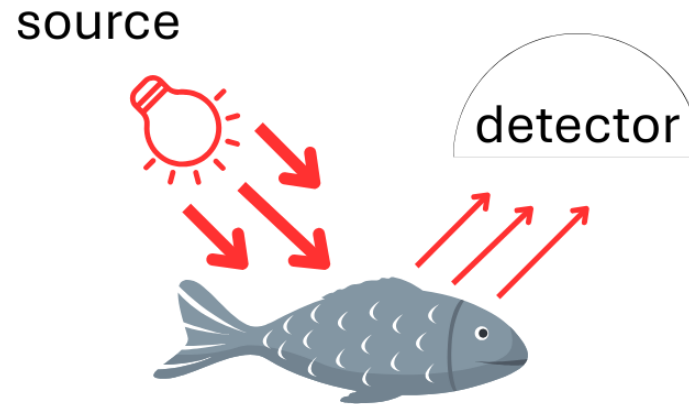


Principles of the method

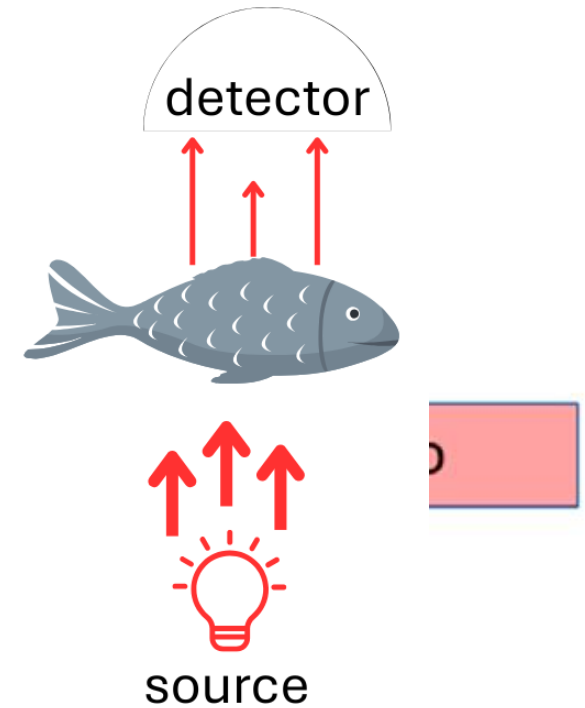
Transmission



Reflectance



Absorbance





SAMPLING

**ACQUISITION OF
REFERENCE DATA**

**SPECTROMETRIC
MEASUREMENT**

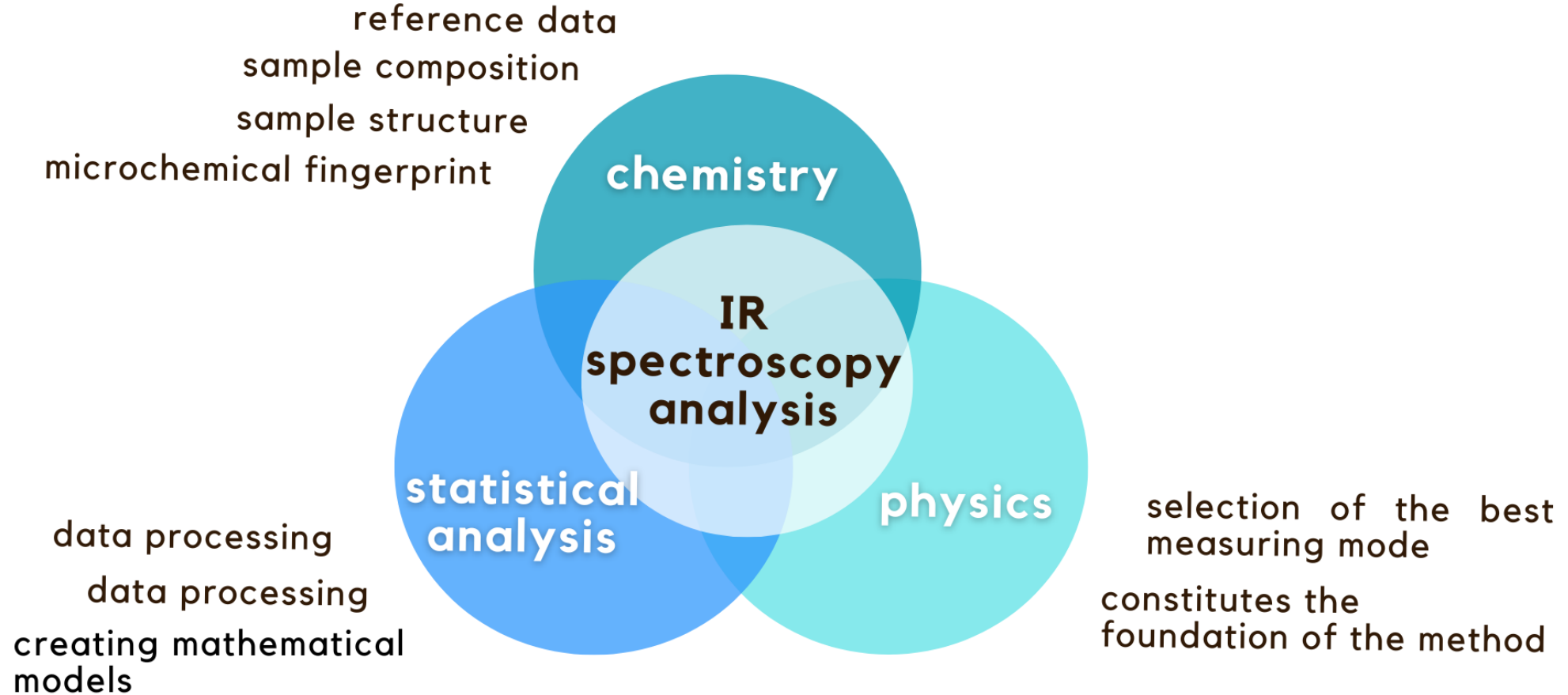
**CHEMOMETRIC
ANALYSIS**

**SPECTROMETRIC
ANALYSIS MODEL**



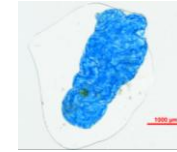
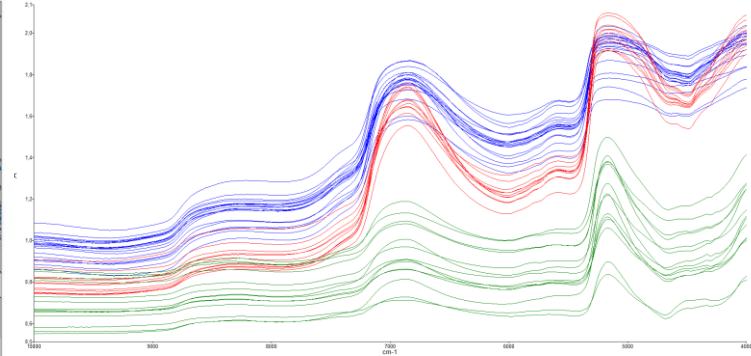
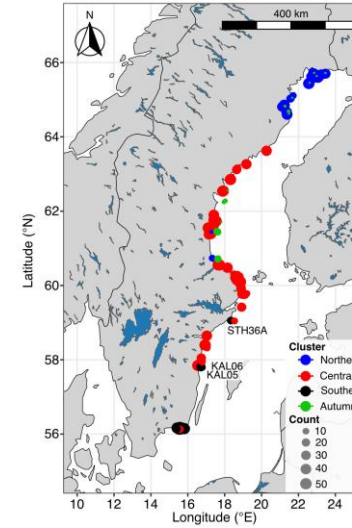
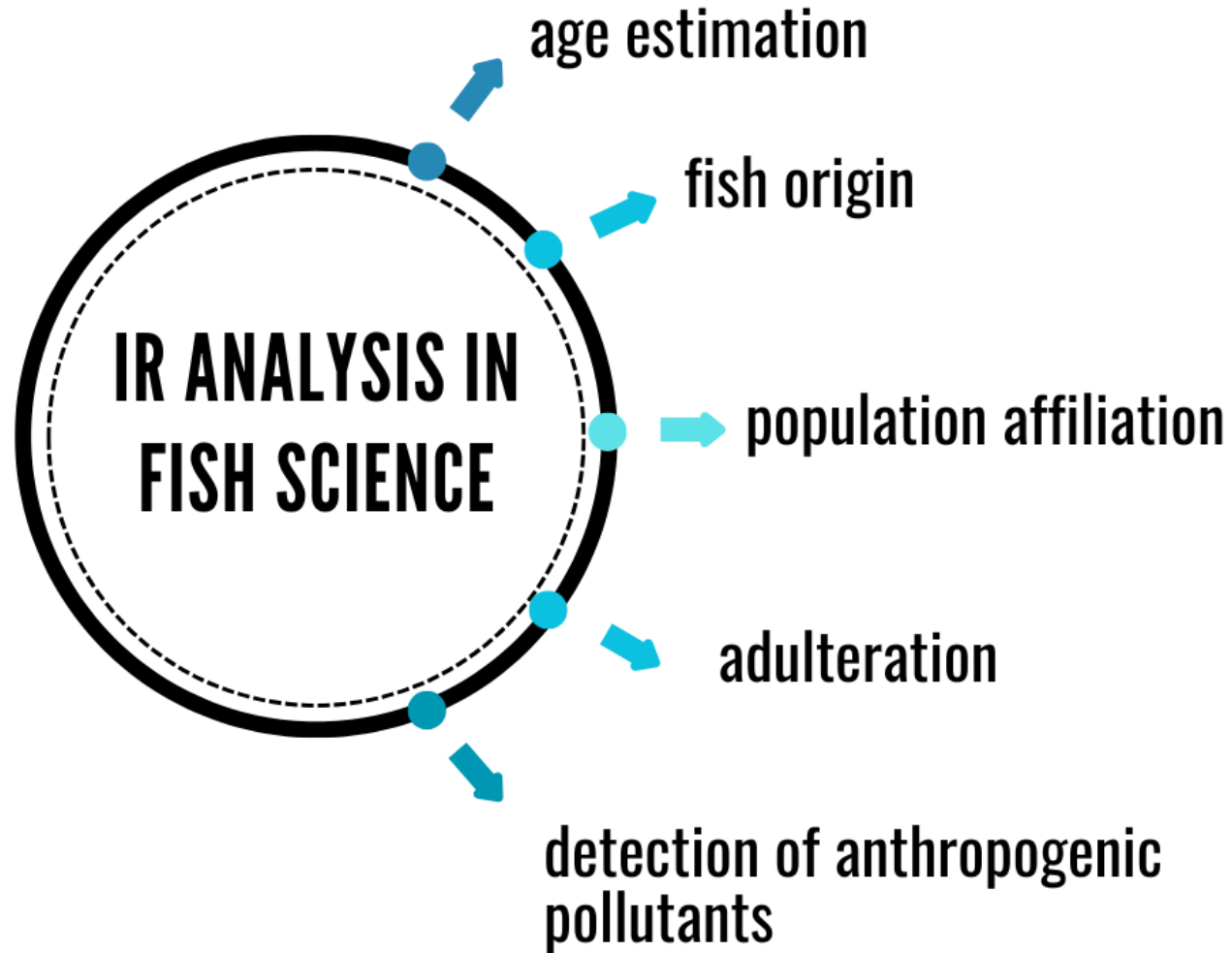


How to interpret data?





Possible application of IR-spectroscopy





Advantages

Non-destructive.

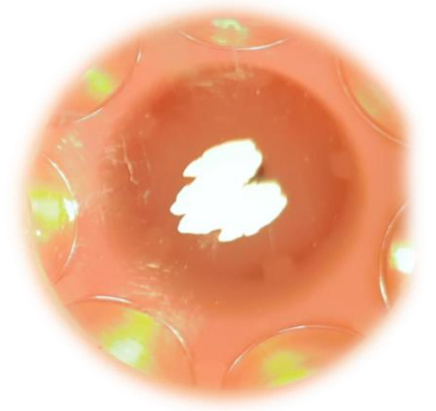
Time-efficient.

Cost-effective.

Suitable for multiple applications

Reduces the use of hazardous substances compared to conventional analysis.

Some spectrometer models are suitable for field applications.





Disadvantages

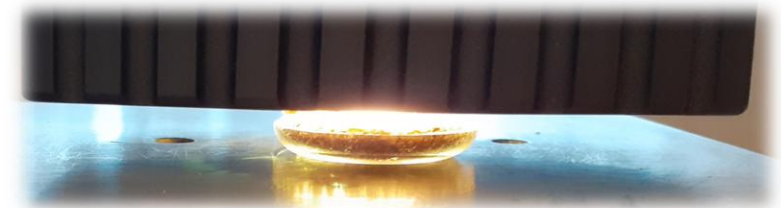
High equipment purchase cost.

Sample size limitation.

Raw data and spectra are difficult to interpret

As an indirect method, it requires reference data.

Requires broader knowledge in the field or work in an interdisciplinary team.





Personal experience



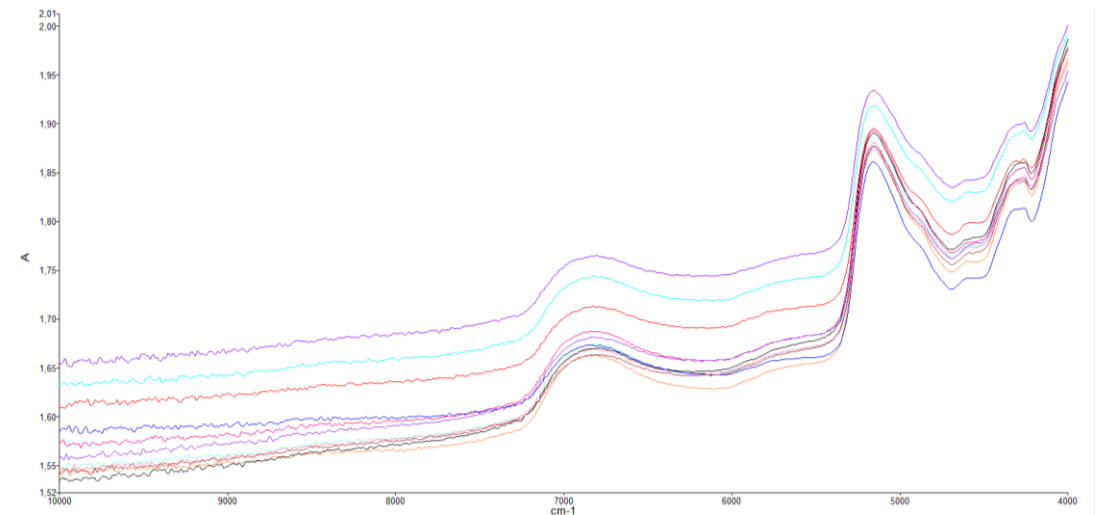
Size does matter.

Different techniques are required for different sample types, analytical aims, and require different spectral interpretations.

The more high-quality reference data you have, the closer the model is to the truth.

Over-processing: the trap of perfection.

Don't hesitate to adapt the method and equipment to your specific needs.





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Thank you for your attention!